The CDC dataset categorizes age using three groups: “Overall,” “50–64 years,” and “65 years or older.” Because the dataset does not provide individual ages or narrower age categories, an under-60 versus 60-years-or-older comparison cannot be directly constructed from the available summarized data. Therefore, the 50–64 versus 65-years-or-older comparison was selected because these are the two age-specific categories directly provided by the dataset.

Research Question

Is there a statistically significant difference in the prevalence of subjective cognitive decline or memory loss between adults aged 50–64 and adults aged 65 years or older?

Null Hypothesis

$\mathbf{H_0}$ - There is no statistically significant difference in the prevalence of subjective cognitive decline or memory loss between adults aged 50–64 and adults aged 65 years or older.

Alternative Hypothesis

$\mathbf{H_1}$ - There is a statistically significant difference in the prevalence of subjective cognitive decline or memory loss between adults aged 50–64 and adults aged 65 years or older.

t-statistic: 17.52039870059076

p-value: 1.4152048496133722e-67

Since the P-value is so small, we **reject the null hypothesis**

Hypothesis test table

Statistic	**Result**
50–64 mean	11.14%
65+ mean	12.20%
Difference	1.06 percentage points
t-statistic	17.52
p-value	1.42×10^{-67}
α	0.05

Decision	Reject H_0

An independent two-sample t-test produced a t-statistic of 17.52 and a p-value of approximately 1.42×10^{-67}. Because the p-value is substantially below the significance level of 0.05, the null hypothesis is rejected. Therefore, the analysis provides strong statistical evidence of a difference in reported SCD prevalence between the 50–64 and 65+ groups. The 65+ group had a mean prevalence approximately 1.06 percentage points higher than the 50–64 group.

Outlier Analysis table

Age group	n	IQR	Lower bound	Upper bound	Outliers	% Outliers
50–64	1,286	3.90	3.05	18.65	54	4.20%
65+	1,260	3.53	4.89	18.99	69	5.48%

Outliers were identified using the 1.5 × IQR rule. The 65+ group contained 69 IQR-defined outliers (5.48%), compared with 54 (4.20%) in the 50–64 group. Thus, the 65+ group had both a greater number and greater proportion of observations classified as outliers.

Before vs After removal chart

Group	Before	After
50–64	11.14%	10.68%
65+	12.20%	11.86%

Excluding IQR-defined outliers reduced the mean SCD prevalence from 11.14% to 10.68% in the 50–64 group and from 12.20% to 11.86% in the 65+ group. The corresponding medians changed only slightly, from 10.70% to 10.50% and from 11.80% to 11.70%, respectively. This indicates that extreme observations had a greater influence on the mean than on the median.

Footnote/outlier table

Age group	IQR outliers	With CDC footnotes	Without footnotes
50–64	54	31 (57.4%)	23 (42.6%)
65+	69	48 (69.6%)	21 (30.4%)

A majority of IQR-defined outliers had associated CDC footnotes. This occurred for 57.4% of outliers in the 50–64 group and 69.6% in the 65+ group. Several extreme observations were associated with regional-estimate warnings or other reporting notes. Therefore, an IQR-defined outlier should not automatically be interpreted as an erroneous observation.

Result From Testing:

Age group	Count	Mean prevalence	Median	SD
50–64 years	1,286	11.14%	10.7%	3.99
65+ years	1,260	12.20%	11.8%	4.21

t-statistic: 17.5204

p-value: 1.415×10^{-67} or 1.4152048496133722e-67

Significance level: α = 0.05

11.14% among adults aged 50–64

12.20% among adults aged 65+

The 65+ group had a higher observed prevalence by about 1.06 percentage points

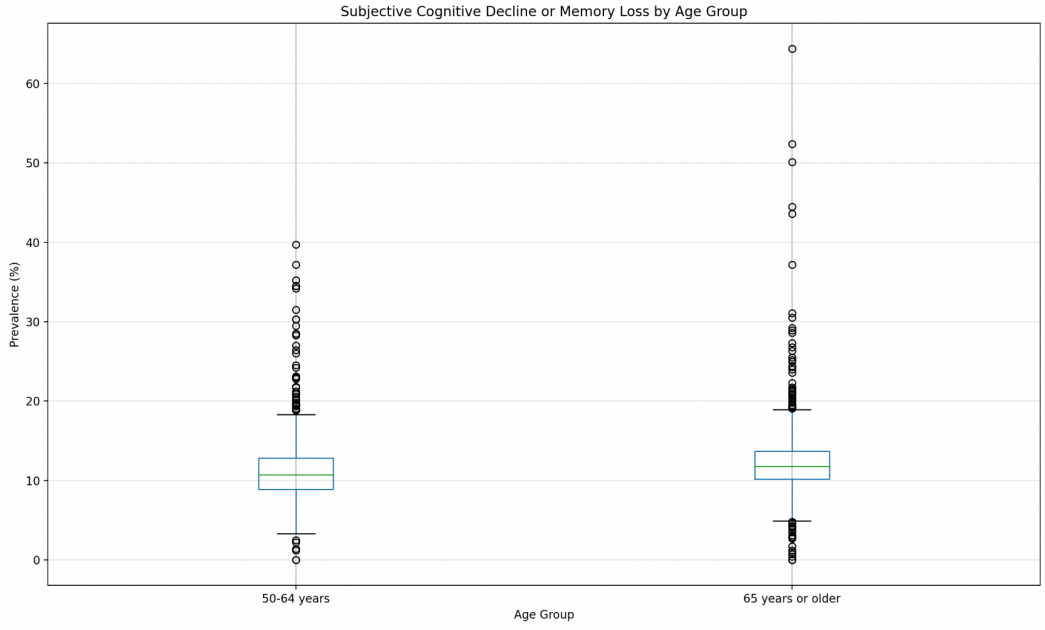


Figure X. Distribution of subjective cognitive decline or memory loss prevalence among adults aged 50–64 years and 65 years or older.
Box plots show the median, interquartile range, and overall distribution of reported prevalence

values for each age group. The 65-years-or-older group has a higher median prevalence than the 50–64-years group, while both groups exhibit several higher-value observations and potential outliers.

Conclusion: Overall, the analysis found a statistically significant difference in reported SCD prevalence between adults aged 50–64 and those aged 65 years or older. The 65+ group had a higher mean and median SCD prevalence and a greater proportion of IQR-defined outliers. Sensitivity analysis showed that removing these outliers reduced the means but had little effect on the medians, and the difference between age groups remained. However, because the CDC dataset consists of aggregated survey estimates rather than individual-level observations, these findings should be interpreted as an association in reported estimates rather than evidence of a causal relationship.

Research Question:

Is there a statistically significant difference in the prevalence of subjective cognitive decline or memory loss between observations with lower and higher obesity prevalence?

Null hypothesis

H_0 - There is no statistically significant difference in the prevalence of subjective cognitive decline or memory loss between observations with lower and higher obesity prevalence.

Alternative hypothesis

H_1 - There is a statistically significant difference in the prevalence of subjective cognitive decline or memory loss between observations with lower and higher obesity prevalence.

Before vs. after outlier removal for obesity

	50–64	**65+**
Mean before	36.696%	28.988%
Mean after	36.664%	29.245%
Median before	36.700%	29.200%
Median after	36.700%	29.300%
N before	465	465
N after	464	455

For the obesity feature, the 50–64 age group had a mean reported prevalence of 36.696%, compared with 28.988% for the 65 years or older group. A Welch two-sample t-test produced a t-statistic of 27.3545 and a p-value below 0.05, so the null hypothesis was rejected, indicating a statistically significant difference between the two age groups.

IQR-based outlier removal identified 1 outlier in the 50–64 group (0.22%) and 10 outliers in the 65+ group (2.15%). After removing these observations, the means were 36.664% and 29.245%, respectively. The mean difference changed from 7.708 to 7.420 percentage points. The Welch t-test remained statistically significant ($t = 27.4223$, $p < 0.05$). Therefore, the statistical significance of the age-group difference was not materially changed by removing the IQR-defined outliers.

Research Question:

Is there a statistically significant difference in reported current-smoking prevalence between adults aged 50–64 and adults aged 65 years or older?

Null Hypothesis

$\mathbf{H_0}$ - There is no statistically significant difference in reported current-smoking prevalence between adults aged 50–64 and adults aged 65 years or older.

Alternative Hypothesis

$\mathbf{H_1}$ - There is a statistically significant difference in reported current-smoking prevalence between adults aged 50–64 and adults aged 65 years or older.

Current Smoking: before vs. after outlier removal

	Before	**After**
50–64 mean	18.348%	18.317%
65+ mean	9.130%	9.112%
Mean difference	9.218 pp	9.205 pp
50–64 N	463	460
65+ N	460	443
t-statistic	43.2526	45.0328
p-value	< 0.05	< 0.05

Removing the IQR-defined outliers had **very little effect on the means**:

- 50–64 changed by only **0.031 percentage points**
- 65+ changed by only **0.018 percentage points**
- The age-group difference changed from **9.218 to 9.205 percentage points**
- The result remained statistically significant.

Interestingly, the t-statistic actually increased from **43.25 to 45.03**, meaning the statistical evidence for a difference did not weaken after removing the outliers.

Descriptive Stats:

Age Group	**N**	**Mean**	**Median**	**SD**	**IQR**	**Outliers**
50–64	463	18.348%	17.900%	4.124	5.300	3 (0.65%)
65+	460	9.130%	9.000%	1.999	2.225	17 (3.70%)

The mean reported current-smoking prevalence was **18.348%** among adults aged 50–64 and **9.130%** among adults aged 65 years or older. Thus, the mean difference was approximately **9.218 percentage points**.